Academic evidence for LLaVA-NeXT in industrial defect detection

LLaVA-NeXT v1.6 Mistral-7B is a defensible choice for Wire DED defect detection fine-tuning, backed by a rapidly growing body of peer-reviewed work. At least 13 papers published between 2023–2025 build on LLaVA architectures for industrial anomaly detection, with the MMAD benchmark (ICLR 2025) establishing LLaVA-NeXT as a standard evaluation baseline. (arXiv) The model's Apache 2.0 license, confirmed LoRA/QLoRA compatibility, and AnyRes high-resolution mechanism make it uniquely practical for EU-based on-premise deployment. However, newer alternatives—particularly **PaliGemma 2** and **Gemma 3**—deserve consideration as they offer competitive or superior fine-tuning performance with smaller footprints. No published work yet combines multimodal LLMs with Wire DED sensor data and melt pool images, placing this thesis at the research frontier.

LLaVA has become the de facto backbone for industrial anomaly detection research

The LLaVA architecture dominates the emerging field of multimodal LLM-based industrial anomaly detection (IAD). The following peer-reviewed and high-venue papers directly build on or evaluate LLaVA for industrial defect detection:

AnomalyGPT (Gu et al., AAAI 2024 Oral) was the first LVLM-based IAD method. Built on the LLaVA-inspired PandaGPT architecture, it uses prompt learning rather than LoRA to avoid catastrophic forgetting on small IAD datasets. (arXiv) (Anomalygpt) On MVTec-AD, it achieves **image-level AUC of 97.4%** and pixel-level AUC of 93.1% in unsupervised settings. (Emergent Mind) Critically, the paper shows that vanilla LLaVA misclassifies nearly all samples as anomalous, demonstrating that domain-specific adaptation is essential. (arXiv) (Anomalygpt)

Triad (Li et al., ICCV 2025) is the strongest direct evidence for choosing LLaVA-NeXT Mistral-7B specifically. This paper uses **LLaVA-1.6-Mistral-7B as its primary backbone**, introducing Expert-Guided Region-of-Interest tokenization that modifies LLaVA-NeXT's AnyRes structure to extract high-resolution anomalous regions. (arXiv) It embeds manufacturing process knowledge via Chain-of-Thought reasoning (CoT-M), achieving state-of-the-art results on MVTec-AD, WFDD, and PCB-Bank benchmarks. (arXiv) Manufacturing process information provides **+1.4% accuracy on MVTec-AD** and +0.9% on WFDD over LLaVA-OneVision. (TheCVF) Fine-tuning takes approximately 3 hours for 0-shot and 5 hours for 1-shot settings. (TheCVF)

VMAD (Deng et al., IEEE Transactions on Automation Science and Engineering, 2024) explicitly uses **LoRA fine-tuning** on LLaVA-7B-v1-1, keeping visual encoders frozen. (arXiv) It introduces Defect-Sensitive Structure Learning and Locality-enhanced Token Compression, (ResearchGate) (arXiv) outperforming Myriad by 3.3% in pixel-level AUC-PR on cross-dataset zero-shot evaluation. (arXiv) This is the clearest precedent for LoRA-based fine-tuning of LLaVA for industrial defect detection.

Diagnostics-LLaVA (Kumar et al., PHM Society 2024) from Hitachi America fine-tunes LLaVA for equipment diagnostics, creating "Automotive-LLaVA" that outperforms both the original LLaVA and mPlugOWL in fault diagnosis, root cause analysis, and repair recommendations. R Discovery phmsociety **FabGPT** (Jiang et al., ICCAD 2024) adapts LLaVA-style architecture with a Vicuna backbone for semiconductor wafer SEM defect detection, using a three-stage training strategy to detect minute defects that GPT-4 and AnomalyGPT miss. arXiv IEEE Xplore **Myriad** (Li et al., arXiv 2023/2024) integrates pre-trained IAD models like PatchCore as "vision experts" into LLaVA, arXiv achieving 61.8% accuracy on PCB-Bank versus AnomalyGPT's 50.1%. arXiv

Additional relevant papers include **EIAD** (Zhang et al., ICME 2025), which introduces a Defect Detection Question Answering dataset from real annotations; arXiv arXiv **IAD-R1** (ICCV 2025), which applies reinforcement learning post-training to LLaVA-1.5 and LLaVA-OneVision, achieving **86.1% average accuracy** as the best open-source IAD model; arXiv and **MAU-GPT** (arXiv 2025), which proposes AMoE-LoRA (Anomaly-aware Mixture-of-Experts LoRA) with a hypernetwork generating sample-specific LoRA weights, outperforming LLaVA-1.5 across all MMAD metrics. arXiv arXiv

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The MMAD benchmark establishes quantitative baselines for industrial contexts

The **MMAD benchmark** (Jiang et al., ICLR 2025) is the first comprehensive evaluation of multimodal LLMs for industrial anomaly detection, GitHub covering **39,672 questions across 8,366 industrial images** in 7 subtasks: OpenReview arXiv anomaly discrimination, defect classification, localization, description, and analysis. OpenReview The Moonlight

Key comparative findings directly relevant to model selection:

- **GPT-4o achieves 74.9%** average accuracy—the best overall, OpenReview +3 but **12% below human experts** and far from industrial requirements arXiv arXiv
- **LLaVA-NeXT** is evaluated as a standard baseline; LLaVA-NeXT-34B ranks among the top-2 open-source models arXiv OpenReview
- LLaVA-1.5 significantly underperforms LLaVA-OneVision and newer models, confirming the architectural improvements in LLaVA-NeXT matter OpenReview
- AnomalyGPT performs worst due to severe overfitting to fixed QA formats— OpenReviewa cautionary tale for narrow fine-tuning arXiv OpenReview
- All models perform dramatically worse on anomaly-specific questions versus general object-related questions OpenReview +2
- **Model scale correlates with performance**: larger open-source models approach commercial capabilities OpenReview

MaViLa (Fan et al., Journal of Manufacturing Systems, 2025) provides a manufacturing-specific comparison showing that general-purpose LLaVA "lacks details" in manufacturing contexts, generating vague descriptions like "a process where a laser is used." ResearchGate Domain-

specific training transforms this into precise process parameter identification and defect recognition. ScienceDirect The paper **From Concept to Manufacturing** (Artificial Intelligence Review, 2025) evaluates LLaVA 1.6 34B alongside GPT-4V across engineering design and manufacturing tasks, Springer finding LLaVA competitive in several areas despite trailing GPT-4V overall.

LoRA fine-tuning of LLaVA is proven across multiple specialized domains

Beyond industrial applications, LoRA/QLoRA fine-tuning of LLaVA is extensively validated across domain-specific visual tasks, establishing the methodology's soundness:

LLaVA-Rad (Zambrano Chaves et al., Nature Communications, 2025) uses LoRA on the LLM component with a three-stage training pipeline for radiology. The result: a **7B model that outperforms GPT-4V and Med-PaLM M (84B)** on chest X-ray report generation, arXiv trainable in one day on 8× A100 GPUs with inference on a single V100. arXiv This is the strongest evidence that LoRA-adapted LLaVA at 7B scale can exceed far larger models on domain-specific visual tasks. **LLaVA-Med** (Li et al., NeurIPS 2023) demonstrates curriculum learning for biomedical adaptation, training in under 15 hours on 8× A100s. arXiv **GeoChat** (Kuckreja et al., 2023) fine-tunes LLaVA-1.5 with LoRA for remote sensing, preserving conversational abilities while adding domain-specific visual understanding. arXiv Mbzuai-oryx **GeoLLaVA** (Elgendy et al., 2024) conducts extensive ablation studies on LoRA rank/scale configurations for LLaVA-NeXT, achieving BERT scores of 0.864 with QLoRA on a single GPU. arXiv **LLaVA-MoLE** (Chen et al., 2024) proposes Mixture-of-LoRA-Experts for handling multi-domain data conflicts during instruction fine-tuning, arXiv outperforming plain LoRA without additional compute. arxiv

A comparative study in **Computers in Biology and Medicine** (2025) benchmarks LoRA and AdaLoRA across five multimodal models including LLaVA-1.5-7B, achieving **90–93% accuracy** on medical VQA with only **15 GB VRAM**. ScienceDirect Typical LoRA configurations across these works use rank 8–64, alpha equal to rank or 2× rank, dropout 0.05, targeting attention projection layers, Medium with learning rates of 1e-4 to 2e-5 over 1–3 epochs.

The Wire DED literature reveals a clear gap for multimodal LLMs

No published paper yet applies multimodal LLMs to Wire DED defect detection combining sensor data and melt pool images—this is the novel contribution space for the thesis. The closest work and the most directly relevant paper identified is:

"In-Context-Learning-Assisted Quality Assessment Vision-Language Models for Metal Additive Manufacturing" (arXiv:2511.05551, November 2025) applies Gemini-2.5-flash and Gemma3:27b VLMs to **wire-laser directed energy deposition (W-LDED)** quality classification using in-context learning. This demonstrates that VLMs can reach classification accuracies comparable to traditional ML models while providing human-interpretable rationale. arXiv

However, it uses proprietary Gemini rather than open-source models and does not combine sensor data with images.

Other additive manufacturing LLM papers include **AdditiveLLM** (Pak et al., Additive Manufacturing Letters, 2025), which fine-tunes text-only LLMs (DistilBERT, SciBERT, Llama, T5) to predict LPBF defect regimes from process parameters, achieving **93% accuracy** arXiv but without vision capabilities. **AMGPT** (Chandrasekhar et al., Additive Manufacturing Letters, 2024) uses Llama2-7B with RAG for AM domain Q&A, outperforming GPT-4 in domain specificity. arXiv ScienceDirect **"Towards Label-Free Defect Detection in Additive Manufacturing"** (Wang et al., Journal of Intelligent Manufacturing, 2026) proposes a dual-classifier semi-supervised VLM approach for AM defect identification that addresses the labeling problem. Springer

The deep learning literature for Wire DED/WAAM defect detection is more mature but relies on traditional architectures. Key works include a **YOLO-attention model** for WAAM surface defects achieving **94.5% mAP at ≥42 fps** ACM Digital Library (Li et al., Robotics and Computer-Integrated Manufacturing, 2023), **U-Net semantic segmentation** for wire-laser DED detecting HAZ and dripping defects Semantic Scholar (Kim et al., IJPEM-Smart Technology, 2025), and **multimodal CNN** frameworks fusing on-axis and off-axis melt pool monitoring for wire DED layer height prediction (Journal of Intelligent Manufacturing, 2025). Multiple sensor fusion papers demonstrate that combining visual and non-visual modalities significantly improves DED defect detection over single-sensor approaches ScienceDirect (Chen et al., ASME IDETC 2023; Gaikwad et al., Additive Manufacturing 2022).

GKN Aerospace operates its AM Centre of Excellence Metal Additive Manufactur… in **Trollhättan, Sweden**, using wire-based DED-LB for titanium and nickel alloy components, with a commissioned "Cell 3"—the world's largest laser-wire DED cell. Metal Additive Manufactur… They have integrated AI-powered anomaly detection trained on high-resolution NXG images, TCT Magazine making this thesis directly industrially aligned.

Competing EU-accessible models that challenge LLaVA-NeXT's position

Several newer models warrant serious consideration as alternatives or ablation baselines:

PaliGemma 2 (Google DeepMind, Gemma License) is the strongest challenger. Available in 3B, 10B, and 28B variants with SigLIP vision encoder and Gemma 2 backbone, it was explicitly designed as a "pre-train then fine-tune" model with built-in object detection and segmentation capabilities. Multi-resolution support (224×224 to 896×896) Hugging Face is valuable for detecting small melt pool defects. Google provides official LoRA fine-tuning notebooks. The 10B variant at 448px offers the best quality-efficiency tradeoff.

Gemma 3 4B-IT (Google DeepMind, Gemma License) uses a "Pan & Scan" adaptive cropping algorithm Hugging Face that automatically zooms into fine details— Label Rer exceptionally useful for manufacturing images with varying defect sizes. At **4.3B parameters**, it fits on a 16GB GPU

with QLoRA while outperforming Gemma-2-27B-IT across benchmarks. Its 128K context window supports processing multiple inspection images. Hugging Face Label Rer

Microsoft Phi-3.5 Vision (MIT License, 4.2B parameters) offers the most permissive licensing and the best performance-per-parameter ratio, scoring MMMU 43.0 NVIDIA Hugging Face versus LLaVA-NeXT Mistral-7B's 35.3, with multi-frame image comparison capability useful for before/after defect analysis. NVIDIA

Idefics2 (HuggingFace M4, Apache 2.0, 8B) uses the same Mistral-7B backbone as LLaVA-NeXT but adds SigLIP vision encoding and native aspect ratio handling. **Pixtral 12B** (Mistral AI, Apache 2.0) from the EU-native Mistral AI achieves MMMU 52.5 Mistral AI but requires more GPU memory. Models excluded due to constraints include all Chinese models (Qwen-VL, InternVL, CogVLM, DeepSeek-VL, MiniCPM-V), Meta Llama Vision (EU-blocked), Idefics3 (Llama 3 backbone), GitHub MiniGPT-4 (Llama backbone), and Fuyu-8B (non-commercial license). Hugging Face

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LLaVA-NeXT v1.6 Mistral-7B: architecture, license, and practical details

The model (HuggingFace: `llava-hf/llava-v1.6-mistral-7b-hf`) comprises three components: a frozen **CLIP-ViT-L/14-336px** vision encoder Cerebras (303.5M parameters), a **2-layer MLP connector** with GELU activation LearnOpenCV (21M parameters), and the **Mistral-7B-Instruct-v0.2** LLM backbone (6.74B parameters), totaling ~7.06B parameters github (~15.1 GB in FP16). Hugging Face

The critical **AnyRes mechanism** splits input images into a grid of up to 2×2 patches at 336×336 each LearnOpenCV (effective maximum 672×672), plus a global downsampled image for context. github Each patch is independently encoded by CLIP, then all features are concatenated and projected through the MLP into the LLM's embedding space. LearnOpenCV TheCVF This delivers **4× more pixels** than LLaVA-1.5's fixed 336×336, directly relevant for detecting fine-grained melt pool anomalies. LLaVA

The original papers to cite are: **"Visual Instruction Tuning"** (Liu et al., NeurIPS 2023 Oral) for the LLaVA architecture, arXiv **"Improved Baselines with Visual Instruction Tuning"** (Liu et al., arXiv:2310.03744, 2023) for LLaVA-1.5, Hugging Face and the **LLaVA-NeXT blog post** (Liu et al., January 2024) for v1.6. LLaVA The Mistral-7B backbone was chosen over Vicuna-7B because it achieves higher MMLU scores LLaVA (60.1 vs 47.1) translating to better multimodal performance across benchmarks LLaVA (MM-Vet: 47.3 vs 43.9; MathVista: 37.7 vs 34.6; TextVQA: 65.7 vs 64.9). Mistral-7B features Grouped-Query Attention Hugging Face Mistral AI and a 32K context window, enabling faster inference and better handling of long visual token sequences.

The **Apache 2.0 license** is confirmed on HuggingFace, inherited from Mistral-7B-Instruct-v0.2. huggingface CLIP-ViT-L/14 uses MIT License. No geographic restrictions exist, making it fully EU-compliant. Mistral AI is itself a French company, Sapling aligning with EU AI ecosystem priorities. One caveat: training data includes GPT-4-generated responses, raising theoretical data

provenance concerns under OpenAI's Terms of Use, though enforcement has been ambiguous. GitHub

LoRA/QLoRA fine-tuning is confirmed working via HuggingFace Transformers + PEFT, with official tutorials available. Emergent Mind +2 A typical QLoRA configuration uses 4-bit NF4 quantization, LoRA rank 32, alpha 16, targeting all linear layers while excluding the vision model and multi-modal projector. Hugging Face Training requires approximately **15 GB VRAM** with QLoRA, Medium fitting on a single RTX 3090 or A6000. UBIAI

Conclusion: a well-supported but not unchallenged choice

The evidence strongly supports LLaVA-NeXT Mistral-7B as a defensible base model. **Triad (ICCV 2025) directly validates this exact model** for industrial anomaly detection with manufacturing process reasoning. The MMAD benchmark (ICLR 2025) establishes LLaVA-NeXT as a standard IAD evaluation baseline. OpenReview +3 LoRA fine-tuning is proven to produce domain-specific LLaVA models that outperform far larger general-purpose systems—LLaVA-Rad (7B) beating GPT-4V and Med-PaLM M (84B) is the most compelling evidence that this approach works. arXiv

Three strategic considerations remain. First, **PaliGemma 2 (3B/10B)** may be a stronger fine-tuning base given its purpose-built architecture, Ultralytics multi-resolution support, Hugging Face and built-in detection capabilities—a comparison ablation would strengthen the thesis. Second, the **~1% defect rate** presents a severe class imbalance challenge that GAN-based augmentation (Chung et al., Journal of Intelligent Manufacturing, 2023) arXiv and semi-supervised approaches (Wang et al., 2026) can address. Third, the thesis occupies a genuine **research gap**: no published work combines multimodal LLMs with Wire DED sensor data and melt pool imagery, positioning the contribution as novel at the intersection of two active research fronts—LLM-based IAD and multimodal DED monitoring.